

A.CREATE TABLE

```
CREATE TABLE main (  
    id INT PRIMARY KEY,  
    salary INT  
);
```

B.Insert data into table

```
INSERT INTO main (id, salary)  
VALUES  
(1, 50000),  
(2, 60000),  
(3, 45000),  
(4, 70000),  
(5, 55000);  
SELECT * FROM main;
```

```
mysql> SELECT * FROM main;  
+----+-----+  
| id | salary |  
+----+-----+  
| 1  | 50000  |  
| 2  | 60000  |  
| 3  | 45000  |  
| 4  | 70000  |  
| 5  | 55000  |  
+----+-----+  
5 rows in set (0.00 sec)
```

1.Before insert Triggers

```
DELIMITER //
```

```
CREATE TRIGGER t_before_insert  
BEFORE INSERT ON main  
FOR EACH ROW  
BEGIN  
    IF NEW.salary < 0 THEN  
        SET NEW.salary = 0;  
    END IF;  
END //
```

```
DELIMITER ;
```

```
INSERT INTO main VALUES (8, -5000);
```

```
SELECT * FROM main;
```

```
mysql> SELECT * FROM main;  
+----+-----+  
| id | salary |  
+----+-----+  
| 1  | 50000  |  
| 2  | 60000  |  
| 3  | 45000  |  
| 4  | 70000  |  
| 5  | 55000  |  
| 8  | 0       |  
+----+-----+  
6 rows in set (0.00 sec)
```

2.After insert

DELIMITER //

```
CREATE TRIGGER t_after_insert  
AFTER INSERT ON main  
FOR EACH ROW  
BEGIN  
    INSERT INTO backup(id, salary)  
    VALUES (NEW.id, NEW.salary);  
END //
```

DELIMITER ;

```
INSERT INTO main VALUES (11, 5000);  
INSERT INTO main VALUES (12, 10000);  
SELECT * FROM main;
```

```
mysql> SELECT * FROM main;  
+-----+-----+  
| id | salary |  
+-----+-----+  
| 1 | 50000 |  
| 2 | 60000 |  
| 3 | 45000 |  
| 4 | 70000 |  
| 5 | 55000 |  
| 8 | 0 |  
| 11 | 5000 |  
| 12 | 10000 |  
+-----+-----+  
8 rows in set (0.00 sec)
```

```
SELECT * FROM backup;
```

```
mysql> SELECT * FROM backup;  
+-----+-----+  
| id | salary |  
+-----+-----+  
| 11 | 5000 |  
| 12 | 10000 |  
+-----+-----+  
2 rows in set (0.00 sec)
```

3.Before update

```
DELIMITER //
```

```
CREATE TRIGGER before_update_salary
BEFORE UPDATE ON employees
FOR EACH ROW
BEGIN
    IF NEW.salary < 5000 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Salary cannot be less than 5000';
    END IF;
END;
//
DELIMITER ;
```

Check output:

Valid: UPDATE employees SET salary = 8000 WHERE id = 1;

```
mysql> SELECT * FROM employees;
+----+-----+-----+-----+
| id | name      | department | salary |
+----+-----+-----+-----+
| 1  | Prajwal   | HR         | 8000   |
| 2  | Prafull   | IT         | 50000  |
| 3  | Priyanshu | Finance    | 40000  |
| 4  | Ravi      | HR         | 20000  |
| 5  | Simran    | IT         | 60000  |
| 6  | Prajwal   | IT         | 90000  |
+----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Invalid: UPDATE employees SET salary = 3000 WHERE id = 1;

```
mysql> UPDATE employees SET salary = 3000 WHERE id = 1;
ERROR 1644 (45000): Salary cannot be less than 5000
mysql>
```

4. After update

```
DELIMITER //
create trigger after_u
after update on main
for each row
begin
    insert into backup(id,salary)
    values(OLD.id, OLD.salary);
    insert into update_log(id, old_salary, new_salary) values (OLD.id, OLD.salary, NEW.salary);
end;
//
DELIMITER ;
```

SELECT * FROM main;

```
mysql> SELECT * FROM backup;
+-----+-----+
| id    | salary |
+-----+-----+
| 11    | 5000   |
| 12    | 10000  |
| 1     | 50000  |
+-----+-----+
3 rows in set (0.00 sec)
```

SELECT * FROM update_log;

```
mysql> SELECT * FROM update_log;
+-----+-----+-----+
| id    | old_salary | new_salary |
+-----+-----+-----+
| 1     | 50000     | 8000      |
+-----+-----+-----+
1 row in set (0.00 sec)
```

5. Before delete

DELIMITER //

create trigger before_delet_main

before delete on main

for each row

begin

insert into backup(id,salary) values (old.id,old.salary);

end;

//

DELETE FROM main WHERE id = 1;

SELECT * FROM backup;

```
mysql> SELECT * FROM main;p
+-----+-----+
| id | salary |
+-----+-----+
| 2  | 60000  |
| 3  | 45000  |
| 4  | 70000  |
| 5  | 55000  |
| 8  | 0       |
| 11 | 5000   |
| 12 | 10000  |
+-----+-----+
7 rows in set (0.00 sec)
```

6. after delete

DELIMITER //

```
CREATE TRIGGER after_delete_main
AFTER DELETE ON main
FOR EACH ROW
BEGIN
    INSERT INTO backup(id, salary)
    VALUES (OLD.id, OLD.salary);
END //
```

DELIMITER ;

```
INSERT INTO main VALUES (1, 5000);
DELETE FROM main WHERE id = 1;
SELECT * FROM backup;
```

mysql> SELECT * FROM backup;

id	salary
11	5000
12	10000
1	50000
1	8000
1	5000
1	5000
2	60000
3	45000
4	70000
5	55000
8	0
11	5000
12	10000
1	5000

14 rows in set (0.00 sec)